

Name: I Forgot Again Team **Date:** 5/25/24 **Class:** Engineering

MyDesign Portfolio

Element K: Reflection on the Design Project

Engineering Design Process Summary (*K1, K2*)

We introduced our problem in element A, broadly describing the overall issue. To complete Element B, we identified previous designs and elaborated on why they failed to solve the problem. Then, in element C, we described our design requirements. For Element D we were able to generate ideas and designs for possible solutions, sketch out drawings, use a decision matrix and a Ghant chart, and list iterations of our design. After narrowing down our prototype options in element D, we asked for feedback from certified STEM experts to help us select our design in element E. Next, we ensured that our prototype fit all the requirements mentioned in element C by comparing it to a table we created in element F. Element G involved listing our construction methods for our prototype and iteration, a CAD drawing, and testing attributes. Element H made us complete testing, testing descriptions, requirements, and details while Element I has the test results. In element J we gathered feedback from stakeholders and field experts to determine what we could improve or do better next time we started a project using the engineering design process.

For each project step, explain: What went well? What didn't go well? Why?

For Elements A, B, and C, our biggest strength was describing the overall problem, mentioning the issue's ins and outs, and providing context as to why and how it affected others. At first, we had difficulty defining the problem because we had trouble collecting our thoughts. All it took was some effective brainstorming and we were back on track. Next, in Element D, we were successful in creating sketches and visualizations of our prototype, however, we had a hard time defining our design requirements through costs and definitions because we still needed to brainstorm. For Element E, we were able to take our design principles and organize them but we had trouble finding real measurements for the requirements. In Element F, creating descriptions for our design requirements was easy, but we needed to work on our measurements as a group. Additionally, Element G was hard because we had trouble explaining our process even if constructing the prototype was fine. In Element H defining our requirements went well, however, finding the measurements for our

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testing was difficult. Also, in Element I our group was good at performing the tests but had a hard time describing our testing processes. Finally, for Element J, our group had strong feedback but we had trouble explaining our process.

Lessons Learned About the Engineering Design Process (K3)

Anyone trying to attempt to navigate the engineering design process should be aware of the process and what is required for each step so that they can adapt and overcome their challenges. For example, It is important to distinguish Element D which is finding solutions to the problem, and choosing one from Element G which is constructing the prototype. Knowing the difference can be the biggest step in solving your problem. From our experiences, the elements can be confusing or challenging, however, people must look at their problem, their goal, and their solution and apply their vision to each part of the process. Engineers shouldn't just use the elements as a guide but as an interactive part of the process. To wrap up, future engineers should take what they know and apply it to the engineering design process.

Element K: Reflection on the Design Project						
K1. Clear and comprehensive summary. The project designer provides _____ of the project.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	a robust and comprehensive summary of each step	a clear and comprehensive summary of the major steps	a generally clear and adequately-developed summary of most steps	a sparse summary of some steps	an overly-general summary	no summary
K2. Value judgment. A value judgment is _____ provided for project steps.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	consistently and robustly	consistently (for major steps)	consistently (for major steps, with one or two addressed in only a cursory manner)	partially (one or two addressed in only a cursory manner)	partially (and overly general or superficial)	not attempted
K3. Inclusion of lessons learned. Lessons learned reflections are	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0

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						not attempted, unclear and/or unlikely to be useful to others attempting engineering design in the future (or there is no evidence of lessons learned)
	substantive and are clearly useful to others attempting engineering design in the future	summarized and are clearly useful to others attempting engineering design in the future	summarized and at least most are useful to others attempting engineering design in the future	included in part and some are useful to others attempting engineering design in the future	minimally included and at least one is useful to others attempting engineering design in the future	